Simple SECR workflow

Namkha basic tutorial

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Session 2 objectives

Main goal: Demonstrate the workflow with a simple example where everything "works":

1. Create a single-session `capthist` object from camera and detection files
2. Build habitat masks for model fitting and reporting
3. Fit a set of `secr` models
4. Turn fitted models into density, abundance, maps, and checks

The emphasis is the workflow: what each object means and how the scripts connect.

The simple workflow

Each script saves the main object needed by the next step:

01
camera file
detection file

`capthist`

02
survey area
buffer
covariates

`mask`

04
density model
detection
model

fitted models

05
AICc
abundance
density
maps

Capture histories: script `01_make_capthist.R`

`01_make_capthist.R` creates a `capthist` object for `secr`. This contains:

1. the **trapfile**: detector locations and related covariates (a `traps` object)
2. the **encounter histories**: recorded encounters of animals at detectors (a data frame)

Tasks involved:

- read and clean camera records
- read detection records
- encode effort as trap usage
- attach trap-level covariates
- create and check a `capthist`
- save the object for later scripts

Capture histories: camera data to `traps`

Set file paths and CRS used by the workflow

```
my_crs <- "+proj=utm +zone=44 +datum=WGS84 +units=m +no_defs"

traps_file <- "tutorials/namkha_basic/data/traps.csv"
detections_file <- "tutorials/namkha_basic/data/detections.csv"

output_file <- "tutorials/namkha_basic/output/namkha_basic_secr_inputs_capthist
fig_dir <- "tutorials/namkha_basic/output/fig"
```

Read in the camera location file (intentionally clean but see later!)

```
cameras <- read.csv(traps_file, stringsAsFactors = FALSE)
```

Capture histories: camera data to `traps`

	session	trapID	Latitude	Longitude	x	y	UTM.Zone	start_date
1	1	Camera8	30.23653	81.53307	551291	3345115	44R	17/12/2022
2	1	Camera11	30.20199	81.45040	543352	3341253	44R	20/12/2022
3	1	Camera14	30.16731	81.38865	537422	3337389	44R	21/12/2022
4	1	Camera12	30.18575	81.43230	541617	3339447	44R	13/12/2022
5	1	TPC48	30.25107	81.74866	572025	3346844	44R	10/12/2022
6	1	TPC38	29.97843	81.51820	549990	3316508	44R	30/12/2022
7	1	TPC23	29.98457	81.59782	557668	3317226	44R	29/12/2022
8	1	TPC20	30.07531	81.47713	545984	3327226	44R	07/11/2022

	end_date	effort	Topography	Altitude	Water	Note
1	13/03/2023	86	Cliff	3978	1	Functional
2	07/02/2023	49	Cliff	4061	0	Functional
3	16/03/2023	85	Cliff	3959	0	Functional
4	16/03/2023	93	Cliff	4079	0	Functional
5	01/06/2023	173	Cliff, Valley floor	4549	1	Functional
6	14/04/2023	105	Hill slope	4132	1	Functional
7	12/03/2023	73	Hill slope	3963	0	Functional
8	18/12/2022	41	Hill slope, Ridgeline	4412	0	Functional

Capture histories: camera data to `traps`

Drop malfunctioning or lost cameras

```
cameras <- cameras |> filter(Note == "Functional")
```

Ensure correct date format

```
cameras <- cameras |>
  mutate(
    start_date = as.Date(start_date, format = "%d/%m/%Y"),
    end_date   = as.Date(end_date,   format = "%d/%m/%Y")
  )
```

Define the overall survey period (here from active cameras)

```
survey_start <- min(cameras$start_date, na.rm = TRUE)
survey_end <- max(cameras$end_date, na.rm = TRUE)
```

Capture histories: camera data to `traps`

Create the `secr` `traps` object

```
traps <- read.traps(
  data = traps_df |> dplyr::select(trapID, x, y, effort),
  detector = "count",
  trapID = "trapID",
  binary.usage = FALSE
)
```

Add covariates that might influence detectability of animals

```
covariates(traps)$Topography    <- traps_df$Topography
covariates(traps)$cliff         <- traps_df$cliff
covariates(traps)$hill          <- traps_df$hill
covariates(traps)$valley_stream <- traps_df$valley_stream
covariates(traps)$Water         <- traps_df$Water
covariates(traps)$Altitude      <- traps_df$Altitude
```

Capture histories: detection data to `capthist`

Read in the detection data (clean; see later for more realistic examples!)

```
capts_raw <- read.csv(detections_file, stringsAsFactors = FALSE)
```

	session	animalID	occasion	date	trapID	sex
1	1	SL1	1	02/16/2023	TPC37	Female
2	1	SL10	1	03/04/2023	TPC14	Female
3	1	SL10	1	05/13/2023	TPC14	Female
4	1	SL11	1	12/18/2022	TPC31	Female
5	1	SL12	1	02/09/2023	TPC25	Male
6	1	SL12	1	11/13/2022	TPC32	Male
7	1	SL13	1	03/04/2023	NRCID296	Female
8	1	SL14	1	03/05/2023	Camera10	Male
9	1	SL14	1	12/10/2022	TPC11	Male
10	1	SL14	1	04/18/2023	TPC11	Male
11	1	SL14	1	01/22/2023	TPC32	Male
12	1	SL15	1	01/12/2023	TPC41	Male

Capture histories: detection data to `capthist`

Only keep detections in the survey period

```
capts_raw <- capts_raw |>
  mutate(date = as.Date(date, format = "%m/%d/%Y")) |>
  filter(date >= survey_start, date <= survey_end)
```

Keep variables used by `secr`

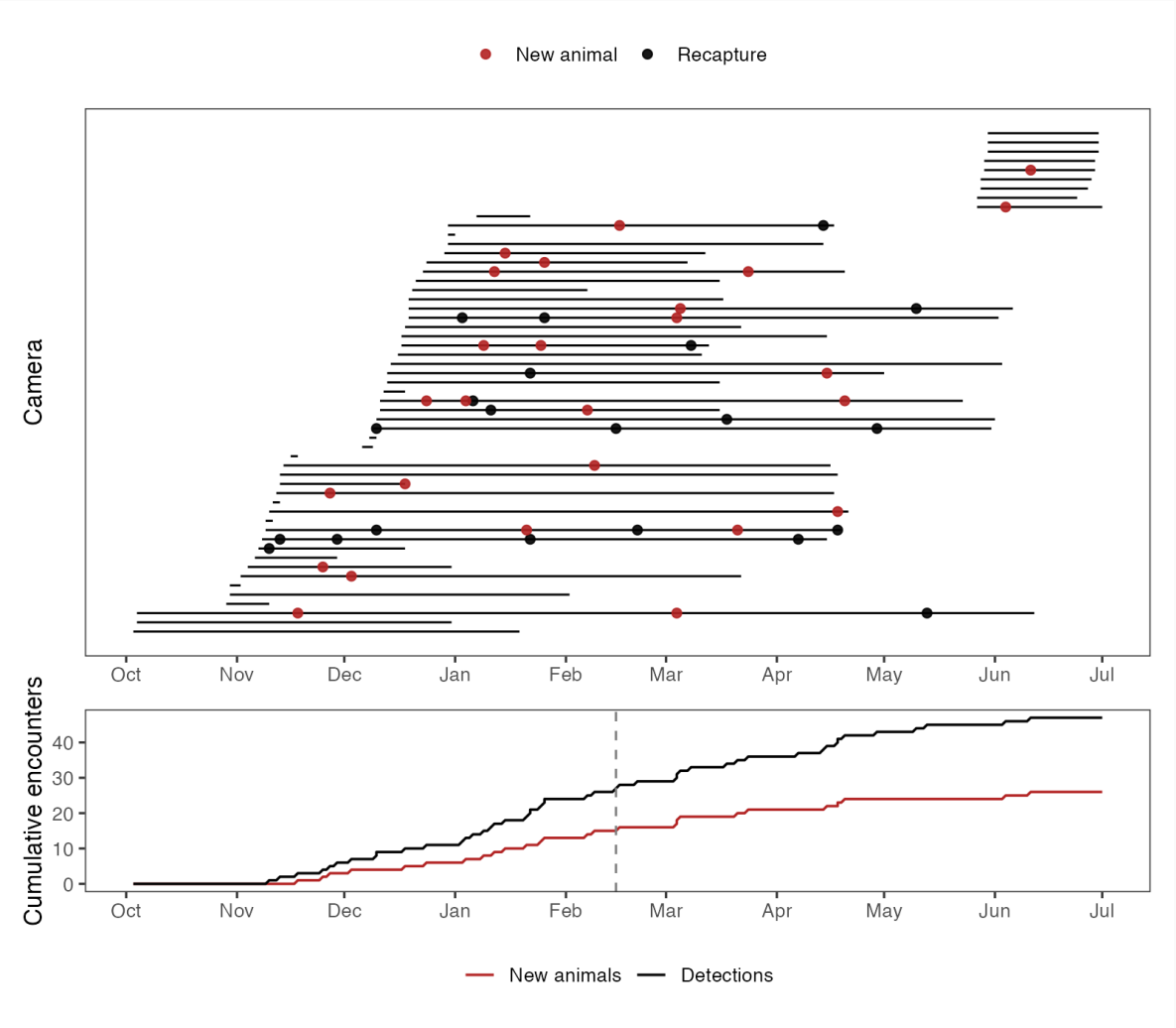
```
capts <- capts_raw |>
  dplyr::select(session, animalID, occasion, trapID, sex)
```

Create and inspect the `secr` `capthist` object:

```
ch <- make.capthist(captures = capts, traps = traps)
verify(ch)
summary(ch, terse = TRUE)
```

Capture histories: descriptive checks

Before fitting models, check that effort and detections are coherent:



- do detections occur during active camera periods?
- new individuals distributed over cameras?
- do new individuals keep appearing late in the survey?

Habitat masks: script `02_make_masks.R`

`02_make_masks.R` creates habitat masks (class `mask`) for fitting and prediction.

Key idea:

- Different masks for model fitting and model prediction
- Model fitting mask needs to include all AC locations animals could be detected from
- Often want to extrapolate to wider area
- *But* where these two masks intersect, useful to share mask points

Implementation:

- Create polygons for (1) close to detectors; (2) extrapolation region
- Find union of these two polygons
- Make mask (points) in this region
- Subset this to form model fitting and prediction masks

Habitat masks: script 02

Tasks involved:

- read the Namkha boundary (extrapolation region)
- build a trap buffer
- create a regular grid of mask points
- add spatial covariates to the mask
- save separate masks for fitting and reporting

Habitat masks: define the region

The full region is the union of the survey boundary and the trap buffer:

```
mask_spacing <- 1500
mask_buffer  <- 25000
```

Aside: How to choose buffer and spacing?

- Get a rough estimate of σ from capthist
- Rough guesses for buffer $= 4\sigma$, spacing $= 0.6\sigma$
- Bigger buffers and small spacings are safer but increase run times

```
qnd <- autoini(capthist = ch, mask = make.mask(traps = traps,
                                               buffer = 75000,
                                               spacing = 1000))
qnd$sigma * 4    # buffer
qnd$sigma * 0.6  # spacing
```

```
Rough buffer:  23704.84

Rough spacing:  3555.726
```

Habitat masks: define the region

Read in extrapolation region polygon

```
survey_area <- st_read("tutorials/namkha_basic/data/
                       survey_area/Namkha_RM.shp",
                       quiet = TRUE) |>
  st_zm() |>
  st_transform(my_crs) |>
  st_geometry()
```

Make traps buffer polygon

```
traps_buffer <- traps_sf |>
  st_buffer(mask_buffer)
```

Full region is the union of the two polygons

```
full_region <- traps_buffer |>
  st_union() |>
  st_geometry()
```

Habitat masks: create the mask

Build a regular grid

```
mask_grid <- st_make_grid(
  # add a small buffer so don't lose points on boundary
  st_buffer(full_region, 10000),
  cellsize = c(mask_spacing, mask_spacing),
  what = "centers"
)
```

Keep points inside the region

```
mask_points <- mask_grid[full_region] |>
  st_as_sf() |>
  cbind(st_coordinates(.)) |>
  st_drop_geometry() |>
  dplyr::select(x = X, y = Y)
```

Create and inspect the `secr` `mask` object

```
mask <- read.mask(data = mask_points, spacing = mask_spacing)
```

Habitat masks: add spatial covariates

Density covariates added to mask because they affect possible AC locations.

Easiest to read in as raster or shape file

```
tri <- rast("tutorials/namkha_basic/data/spatial_covs/TRI_Namkha.tif")
# remember to project to same CRS as mask
tri <- project(tri, y = my_crs)
names(tri) <- "tri"
mask <- addCovariates(mask, tri)
```

May need some preprocessing (more later on covariates)

```
covariates(mask)$d2hydro <- as.numeric(st_distance(mask_sf, hydro))
```

- `mask_model` is used for `secr.fit()`
- `mask_survey_area` is used for abundance and reporting
- these can but don't have to differ

Model fitting: script 04

`04_model_fitting.R` fits a set of candidate models.

- start with the null model
- add density covariates such as `std_tri` or `std_d2hydro`
- add detector covariates such as `valley_stream`
- use AIC for model selection

Model fitting: adding covariates

Fitting the null model:

```
m0 <- secr.fit(ch,
  detectfn = "HHN",
  mask = mask_model,
  model = list(D ~ 1, lambda0 ~ 1, sigma ~ 1)
)
```

`m4` has covariates on density and detection

```
m4 <- secr.fit(ch,
  detectfn = "HHN",
  mask = mask_model,
  model = list(D ~ std_tri, lambda0 ~ valley_stream, sigma ~ 1),
  start = m0
)
```

Model fitting: adding covariates

Use AICc or AIC to compare the candidate set after fitting

```
AIC(m0, m1, m2, m3, m4, m5, criterion = "AICc")
```

```
                                                                 model
m3                                    D~1 lambda0~valley_stream sigma~1
m4                                D~std_tri lambda0~valley_stream sigma~1
m0                                                D~1 lambda0~1 sigma~1
m5 D~std_tri + std_d2hydro lambda0~cliff + valley_stream sigma~1
m1                                          D~std_tri lambda0~1 sigma~1
m2                                       D~std_d2hydro lambda0~1 sigma~1
            detectfn npar    logLik     AIC    AICc dAICc AICcwt
m3 hazard halfnormal    4 -106.0014 220.003 221.908 0.000 0.6465
m4 hazard halfnormal    5 -105.9684 221.937 224.937 3.029 0.1422
m0 hazard halfnormal    3 -109.1400 224.280 225.371 3.463 0.1145
m5 hazard halfnormal    7 -103.7172 221.434 227.657 5.749 0.0365
m1 hazard halfnormal    4 -109.0375 226.075 227.980 6.072 0.0311
m2 hazard halfnormal    4 -109.0967 226.193 228.098 6.190 0.0293
```

Interpreting the null model – m_0

Type the name of the model for a model summary

```
secr.fit(capthist = ch, model = list(D ~ 1, lambda0 ~ 1, sigma ~
    1), mask = mask_model, detectfn = "HHN", verify = FALSE,
    trace = FALSE)
secr 5.4.2, 11:45:50 28 Apr 2026

Detector type       count
Detector number     55
Average spacing     2504.368 m
x-range             534489 576620 m
y-range             3316462 3360989 m

Usage range by occasion
      1
min   2
...
```

Interpreting the null model – m_0

Type the name of the model for a model summary

```
...
max 251

N animals        :  26
N detections     :  47
N occasions      :  1
Count model      :  Poisson
Mask area        :  667350 ha

Model            :  D~1 lambda0~1 sigma~1
Fixed (real)     :  none
Detection fn     :  hazard halfnormal
Distribution     :  poisson
N parameters     :  3
Log likelihood   :  -109.14
AIC              :  224.2801
AICc             :  225.371
...
```

Interpreting the null model – m_0

Type the name of the model for a model summary

```
...

Beta parameters (coefficients)
             beta   SE.beta       lcl       ucl
D       -8.718803 0.2520262 -9.212766 -8.224841
lambda0 -5.965939 0.2929625 -6.540134 -5.391743
sigma    8.711655 0.1437296  8.429950  8.993359

Variance-covariance matrix of beta parameters
                   D      lambda0       sigma
D        0.063517229 -0.009673521 -0.01379971
lambda0 -0.009673521  0.085827004 -0.02741592
sigma   -0.013799708 -0.027415917  0.02065819

Fitted (real) parameters evaluated at base levels of covariates
        link     estimate  SE.estimate          lcl          ucl
D        log 1.634827e-04 4.186493e-05 9.975776e-05 2.679149e-04
lambda0  log 2.564636e-03 7.677555e-04 1.444294e-03 4.554030e-03
sigma    log 6.073283e+03 8.774382e+02 4.582270e+03 8.049453e+03
...
```

Interpreting the null model – m_0

- `coef()` for model coefficients on the link scale

```
coef(m0)
```

```
             beta   SE.beta       lcl       ucl
D       -8.718803 0.2520262 -9.212766 -8.224841
lambda0 -5.965939 0.2929625 -6.540134 -5.391743
sigma    8.711655 0.1437296  8.429950  8.993359
```

- `predict()` for model coefficients on the response scale

```
predict(m0)
```

```
        link     estimate  SE.estimate          lcl          ucl
D        log 1.634827e-04 4.186493e-05 9.975776e-05 2.679149e-04
lambda0  log 2.564636e-03 7.677555e-04 1.444294e-03 4.554030e-03
sigma    log 6.073283e+03 8.774382e+02 4.582270e+03 8.049453e+03
```

Interpreting a covariate model – m_4

- `coef()` for model coefficients on the link scale

```
coef(m4)
```

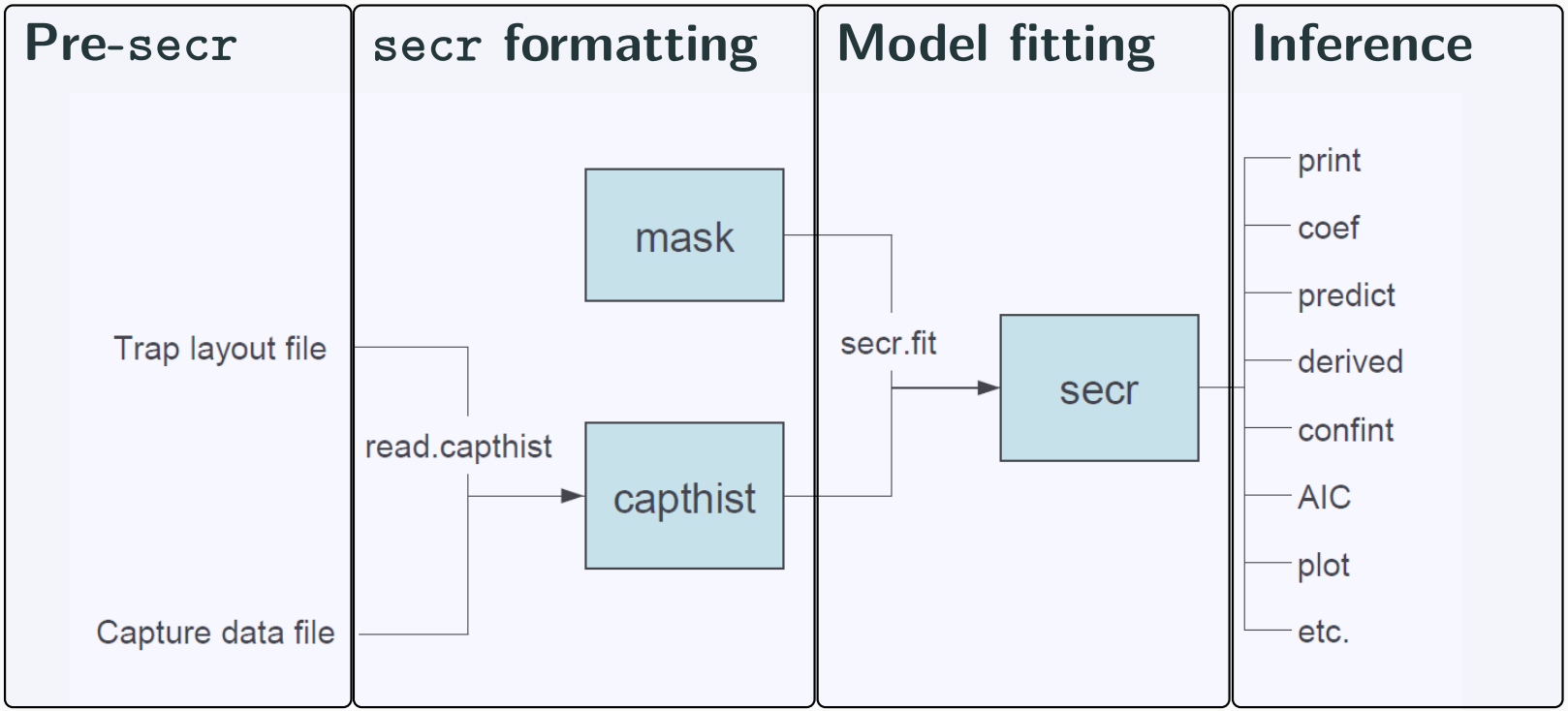
```
                            beta   SE.beta       lcl         ucl
D                     -8.6926618 0.2604923 -9.203217 -8.18210629
D.std_tri             -0.1021571 0.3950087 -0.876360  0.67204584
lambda0               -5.8202128 0.2963861 -6.401119 -5.23930669
lambda0.valley_stream -1.3093661 0.6198281 -2.524207 -0.09452529
sigma                  8.7232095 0.1455221  8.437991  9.00842768
```

- `predict()` for model coefficients on the response scale

```
predict(m4)
```

```
        link     estimate  SE.estimate          lcl          ucl
D        log 1.678127e-04 4.446608e-05 1.007148e-04 2.796124e-04
lambda0  log 2.966974e-03 8.990400e-04 1.659699e-03 5.303933e-03
sigma    log 6.143866e+03 8.988229e+02 4.619267e+03 8.171663e+03
```

secr workflow



Post processing functions in `secr`

A bunch of helper functions for working with model output:

- extract link-scale estimates & associated uncertainty: `coef()`
- extract real-scale estimates & associated uncertainty: `predict()`
- compute the population size for the area: `region.N()`
- plot detection function based on estimates: `plot()`
- plot estimated AC locations: `fxi.contour()`
- many more

Estimation and reporting: script 05

`05_model_results.R` turns fitted models into common outputs.

Tasks involved:

- compare models with AICc
- summarise abundance and density over `mask_survey_area`
- choose the best model for mapping by AICc
- map the survey region and predicted density
- inspect covariate effects on density and detection

Estimation and reporting: build the summary table

Choose models you want to report results for

```
fitted_models <- list(m0 = m0, m1 = m1, m2 = m2,
                      m3 = m3, m4 = m4, m5 = m5)
```

Choose area you want to report results for:

```
extrap_mask <- mask_survey_area
```

Extracting abundance for all chosen models (`extract_abundance` is a wrapper function for `secr`'s `region.N`):

```
abundance_table <- lapply(names(fitted_models),
                          extract_abundance,
                          abundance = c("E.N"))
```

Note: `abundance = c("E.N")` for expected abundance, `abundance = c("R.N")` for realised abundance

Estimation and reporting: build the summary table

Reshape into nice format:

```
abundance_table <- abundance_table |> bind_rows() |>
  mutate(D = estimate / survey_area_km2,
         Dlcl = lcl / survey_area_km2, Ducl = ucl / survey_area_km2,
         CV = 100 * SE.estimate / estimate)
```

```
  modelname estimate SE.estimate      lcl      ucl          D        Dlcl
1        m0 39.43203   10.097822 24.06157 64.62109 0.01634827 0.009975776
2        m1 38.94529    9.789949 23.97462 63.26422 0.01614647 0.009939726
3        m2 39.30222    9.960437 24.10148 64.09002 0.01629445 0.009992320
4        m3 39.80810   10.261072 24.21485 65.44269 0.01650419 0.010039323
5        m4 40.35744   10.540280 24.39325 66.76941 0.01673194 0.010113290
6        m5 42.00740   11.312175 25.00953 70.55795 0.01741600 0.010368794
        Ducl       CV
1 0.02679149 25.60817
2 0.02622895 25.13770
3 0.02657132 25.34320
4 0.02713213 25.77634
5 0.02768218 26.11731
6 0.02925288 26.92901
```

Estimation and reporting: abundance and density

AIC's for chosen models:

```
aic_table <- AIC(m0, m1, m2, m3, m4, m5, criterion = "AICc") |>
  mutate(modelname = row.names(.)) |>
  left_join(abundance_table, by = "modelname")
```

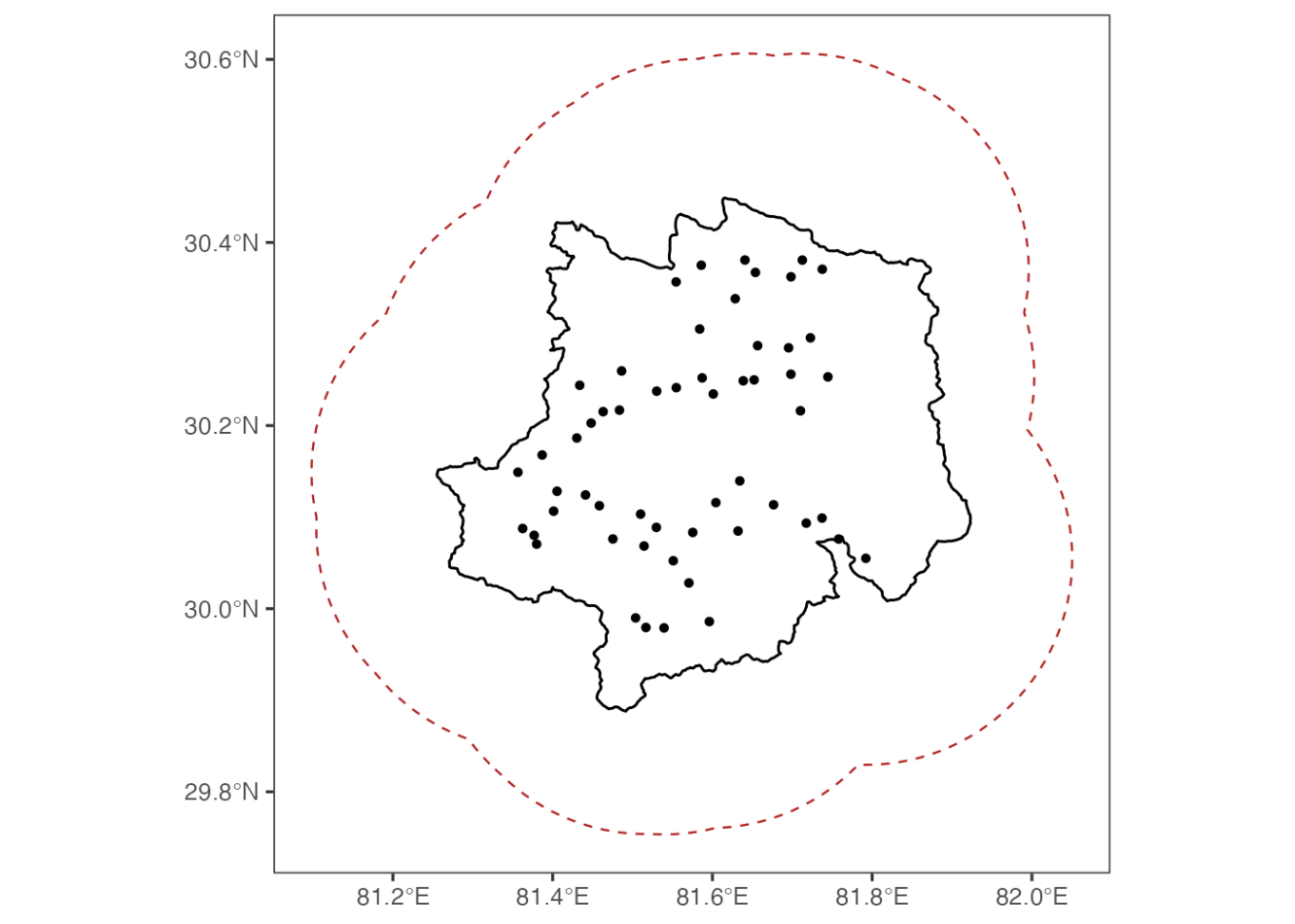
Best model is one with lowest AIC:

```
best_model_name <- aic_table$modelname[which.min(aic_table$AICc)]
best_model <- fitted_models[[best_model_name]]
```

Save results:

```
write.csv(aic_table |>
            select(modelname, model, AIC, AICc, dAICc, AICcwt,
                   estimate, lcl, ucl, D, Dlcl, Ducl, CV),
          file = "output/namkha_basic_results_abundance_AIC.csv",
          row.names = FALSE)
```

Estimation and reporting: survey region



Survey boundary, camera locations, and the trap-buffer region used in the workflow.

Estimation and reporting: predicted density

Predicted density at each point in the extrapolation mask:

```
pred_surface <- predictDsurface(best_model,
                                mask = extrap_mask,
                                cl.D = FALSE)
```

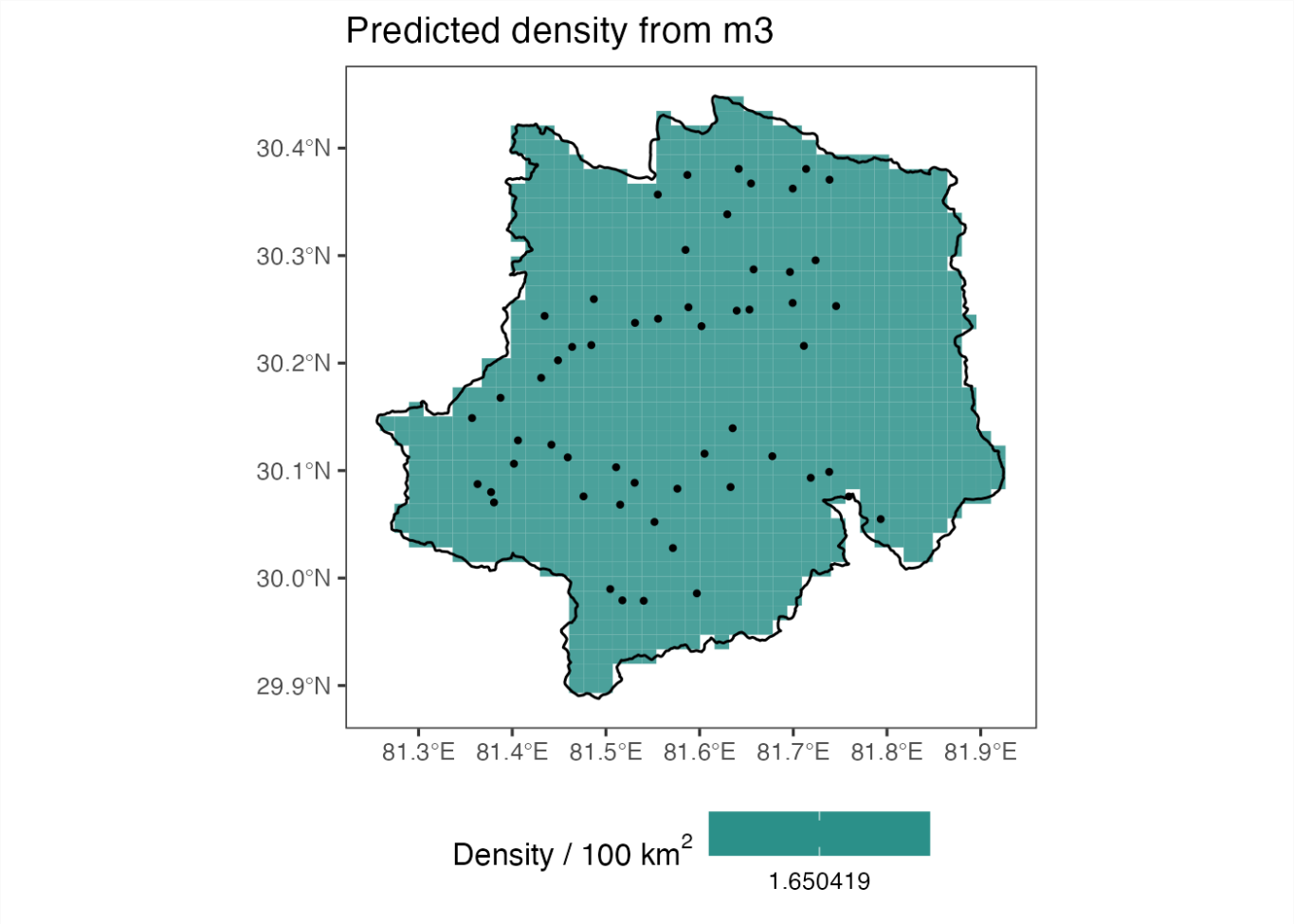
Create grid cells centered on mask points for plotting:

```
density_cells <- st_as_sf(data.frame(extrap_mask),
                          coords = c("x", "y"), crs = my_crs)
density_cells <- st_buffer(
  density_cells,
  dist = attr(extrap_mask, "spacing") / 2,
  endCapStyle = "SQUARE",
  nQuadSegs = 1
)
```

Add in the densities per 100 km^2:

```
density_cells$D_per_100km2 <- covariates(pred_surface)[, "D.0"] * 10000
```

Estimation and reporting: predicted density



Estimation and reporting: density covariate effects

To show how density changes with one focal covariate, choose a model and covariate

```
covplot_model <- m5
focal_cov <- "std_tri"
focal_cov_full_name <- "Terrain ruggedness index"
```

Would usually choose the AIC-best model but here using `m5` which has two density covariates and two detection covariates

```
                           beta   SE.beta         lcl         ucl
D                    -8.4550097 0.4066118 -9.25195416 -7.6580653
D.std_tri            -0.2652649 0.3984637 -1.04623940  0.5157097
D.std_d2hydro         0.4465780 0.7186844 -0.96201753  1.8551736
lambda0              -6.0508404 0.3335679 -6.70462142 -5.3970594
lambda0.cliff         0.6394071 0.3173501  0.01741238  1.2614018
lambda0.valley_stream -1.3472480 0.6278909 -2.57789158 -0.1166045
sigma                 8.6865962 0.1448344  8.40272591  8.9704664
```

Estimation and reporting: density covariate effects

Create a range of potential values for the focal covariate:

```
newdata <- data.frame(
  focal_values = seq(
    min(covariates(extrap_mask)[, focal_cov], na.rm = TRUE),
    max(covariates(extrap_mask)[, focal_cov], na.rm = TRUE),
    length.out = min(500, nrow(extrap_mask))
  )
)
names(newdata)[1] <- focal_cov
```

Other covariates are set to their mean or mode (not shown here)

Estimation and reporting: density covariate effects

Predicting densities at new covariate values is a bit fiddly.

- Use a temporary mask so `predictDsurface` can evaluate the density model.

```
pred_mask <- subset(extrap_mask, subset = 1:nrow(newdata))
```

- Replace covariate values with the ones we want

```
for (covname in names(newdata)) {
  covariates(pred_mask)[, covname] <- newdata[[covname]]
}
```

- Now predict density

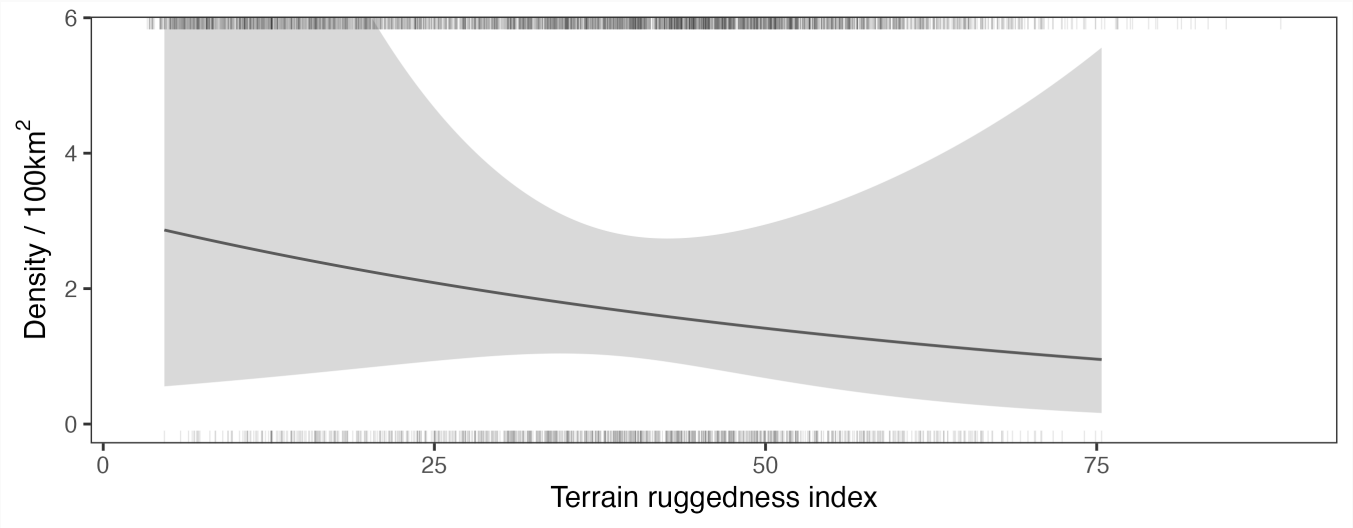
```
tmp <- predictDsurface(covplot_model, mask = pred_mask, cl.D = TRUE)
```

Estimation and reporting: density covariate effects

- Extract observed covariate values on mask and extrapolation region to assess covariate coverage and include this on plot

```
x_surveyed <- data.frame(xt = c(covariates(mask_model)[, plot_cov]))
x_full <- data.frame(xt = c(covariates(extrap_mask)[, plot_cov]))
```

- Finally plot:



Estimation and reporting: encounter rate covariate effects

Identify which covariates appear in the encounter-rate part of the chosen model

```
model_covs <- covplot_model$betanames
lam0_covs <- sub("^lambda0\\.", "",
                 model_covs[grepl("^lambda0\\.", model_covs)])
lam0_covs
```

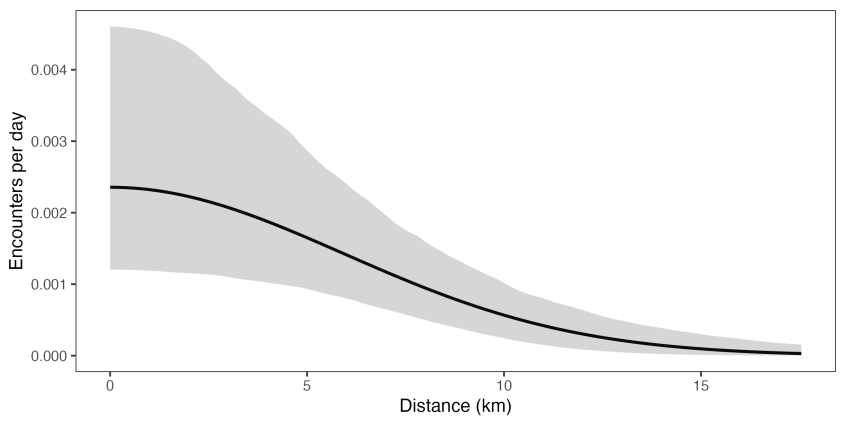
```
[1] "cliff"         "valley_stream"
```

Set possible values for any λ_0 covariates **in same order as** `lam0_covs`

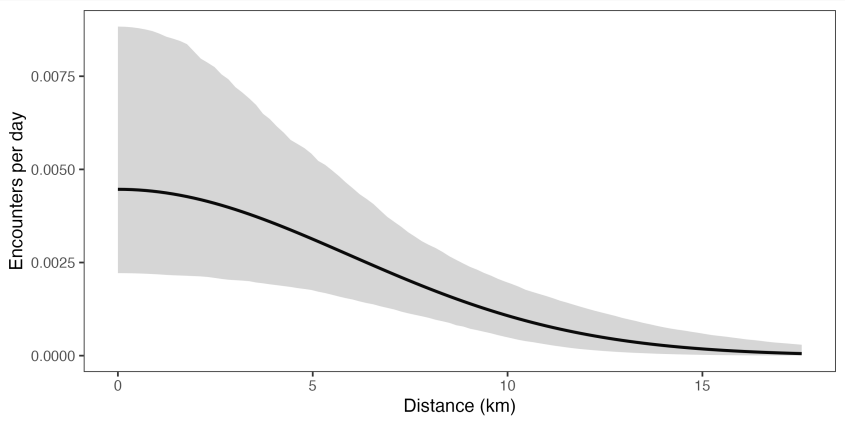
```
lam0_covs_vals <- c(1, 0) # for cliff
lam0_covs_vals <- c(0, 1) # for valley/stream
lam0_covs_vals <- c(0, 0) # for baseline (other)
```

Estimation and reporting: encounter rate covariate effects

Plot for baseline:



Plot for cliff:



Workflow recap

1. Read in camera and detection csv's and create `capthist`
2. Build masks for model fitting and reporting (keep these concepts separate)
3. Fit candidate `secr` models
4. Produce interpretable output: density, abundance, maps, covariate effects

What did we miss?

1. Survey period too long for population closure
2. Barriers to movement
3. Other covariates
4. Sex-dependent parameters
5. Perfectly clean data doesn't exist
6. Probably lots more!

Some of these are covered in the next sections